বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি, কুমিল্লা
BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Mid Term Examination, Spring 2025

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE-225

Course Title: Numerical Methods

Exam Duration: 1.5 Hours

Full Marks: 90

Notes:

a. Figure on the right of each question indicate marks for respective question.
b. **Answer all the questions.** Each question carries 30 marks.
c. Course Outcomes are:

CO1: Recall the basics of accuracy and finite precision, round-off errors, truncation errors, and stability of numerical algorithms.
CO2: Apply appropriate numerical techniques and calculate the computational cost in the fields of science and engineering.
CO3: Formulate numerical techniques for root finding, curve fitting, differentiation, and integration.

1. a. What is Round-off Error? How is it affected during the process of numerical computing? Explain with an example. [CO1→C1, C2] 10

b. A temperature sensor records a reading of 37.8921°C. What is the round-off error if this value is stored using a four-digit mantissa? [CO1→C1, C2] 10

c. Convert 2.75 and 0.125 into 8-bit binary form. Add them and convert the result back to decimal. Check for any conversion error. [CO1→C3] 10

OR

a. A sensor in a lab measures temperature with the following values: 15

37.5000°C	0.002345 mV	5.6789 atm	0.0456000 m	3600.00 s	0.000789 A

i) Identify the number of significant digits in each measurement.
ii) Rank the measurements from highest to lowest precision. [CO1→C5]

15

b. Two floating-point numbers are given in scientific notation as 9.999×10^{38} and 8.000×10^{38}.

i) Add these two numbers following the floating-point addition algorithm, showing the intermediate steps, including alignment of the exponents, addition of the significands, and normalization. [CO1→C3]

ii)Determine if the result causes an overflow and explain why this happens. Recall that the largest positive single-precision floating-point number is approximately 3.4×10^{38}. [CO1→C1]

2. a. Apply the bisection method to find a root of the function $f(x) = x^2-5$ in the interval $[2, 3]$ up to 5 iterations. Verify if the root is accurate up to 3 decimal places. [CO2→C3] 15

b. Given the function $f(x) = x^2-4$ in the interval $[1,3]$, solve the equation for both the bisection method and false position method to find the root up to 4 iterations. Compare the accuracy of the two methods for this function. [CO2→C3] 15

3 a. Write down the condition for convergence of the fixed-point iteration method. Apply the fixed-point iteration method to find the root of the equation $f(x) = 2x^3 - 2x - 5$ correct to four decimal places. [CO2→C3] 15

b. Compute a root for the equation, $f(x) = 3x - cosx - 1$ using the Newton-Raphson method and correct it to three decimal places. [CO2→C3] 15

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Mid Term Examination, Spring 2025

Department of Computer Science and Engineering (CSE)
Level- 2, Term-II
Course Code: MATH 221
Course Title: MATH-III
Credit Hour: 03

Notes:

Time : 1 hr 30 mins

Full Marks: 90

a. Figure on the right of each question indicate marks for respective question.
b. Answer any **three (03)** out of the following **four (04)** questions.

1. a. Suppose the following data represents the Midterm marks (Math) about 32 students **20**

72, 48, 37, 56, 67, 74, 82, 36, 45, 42, 66, 55, 72, 86, 51, 69, 77, 91, 62, 79, 38, 53, 60, 74, 58, 65, 72, 73, 56, 42, 85, 54

(i) **Determine** arithmetic mean by using short-cut method

(ii) **Visualize** the data by Cumulative frequency polygon.

b. Using the observations of **1(a) show** that $G.M \geq H.M$. **10**

2. a. For any two observations **show** that $S.D. = \frac{1}{2} \times |a - b|$ **10**

b. A frequency distribution table for the Daily wages of 35 labors in a certain factory are given below **20**

Class	11-13	13-15	15-17	17-19	19-21	21-23	23-25
No. of labors	3	4	5	10	6	4	3

(a) **Compute** the semi-interquartile of their daily wages.

(b) **Identify** the dispersion by using co-efficient of variation.

3. a. **Estimate** the median and mode of data set 12, 8, 6, 7, 13, 14, 12, 6, 8, 12, 6, 7 **10**

b. A survey of 80 students has given below in a data table. **Compute** $Q3$, $D4$ and $P25$ from the data set. **20**

Tiffin Cost	51-55	56-60	61-65	66-70	71-75	76-80	81-85
No. of students	14	7	9	10	13	12	15

4. a. Using frequency data table of **Q3(b) express** the highest number of frequency from the Pie-chart (using angle). **10**

b. **Find** the regression coefficient and obtain the lines of regression for the following data **20**

X	1	2	3	4	5	6	7
Y	14	7	9	10	13	12	15

29/05/25
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Mid Term Examination, Spring 2025
Department of Computer Science and Engineering

Level-2 Term-II
Course Code: CSE 221
Course Title: Computer Algorithms and Complexity
Credit Hour: 3.0
Exam Duration: 1 hour 30 Minutes
Full Marks: 90

Notes:

a. Figure on the right of each question indicates marks for respective question.
b. Course Learning outcomes are:
CLO1: Analyze the asymptotic performance and write rigorous correctness proofs for algorithms.
CLO2: Estimate the performance of various algorithms and data structures.
CLO3: Apply important algorithmic design paradigms and methods analysis for problem solving.
CLO4: Device efficient algorithms in common engineering design situations.

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1. a. The Fibonacci series is a sequence where each number is the sum of the two preceding ones, like 0, 1, 1, 2, 3, 5, ... n. Write the pseudocode to generate the Fibonacci series up to n terms. Clearly explain the logic used. **[CO1→C2]** 10

b. Construct a Step Table for the following algorithm. **[CO1→C2]** 10

```
Algorithm Add(a,b,c,m,n)
{
        for i:=1 to m do
            for j:=1 to n do
                c[i, j] := a[i,j] + b[i,j];
}
```

c. Explain how Merge Sort and Quick Sort divide and conquer the input array. Highlight the key differences in their recursive strategies. **[CO1→C2]** 10

OR

What is an algorithm? Briefly explain the essential characteristics of algorithm. **[CO1→C1]**

2. a. Describe how a Min Heap is constructed from the following array [40, 30, 50, 22, 60, 55, 77, 35]. Show the steps. **[CO2→C3]** 10

b. Describe the step-by-step process of the Selection Sort algorithm for sorting an array [8, 4, 6, 9, 2, 3, 1] in ascending order. **[CO2→C3]** 10

c. Provide examples of real-world applications where stacks are used. Discuss how stacks are employed in these scenarios to solve specific problems efficiently. [CO2→C2] 10

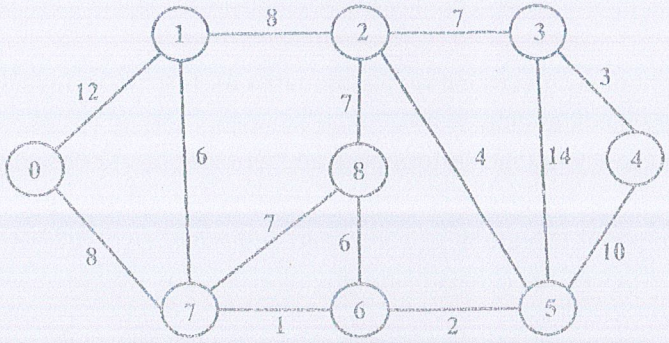
OR

Construct a 5x5 magic square using the numbers 1 to 25. Provide a step-by-step explanation of the method you used. [CO2→C4]

3. a. Write a pseudo code for recursive search of a Binary Search Tree(BST). [CO3→C3] 15

b. Implement the Quick Sort Algorithm for sorting a large dataset. [CO3→C3] 15

OR



Illustrate the step-by-step execution of Prim's algorithm on a given weighted graph, indicating how the MST is constructed. [CO3→C4]

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Mid-Term Examination, Spring 2025

Department of Computer Science and Engineering

Level-2, Term-II

Course Code: GED 221

Course Title: Engineering Economics

Notes: **Time: 1 Hour 30 Minutes**

a. Answer any *03* (*Three*) of the following *04* (*Four*) questions. **Full Marks: 90**

b. Each question carries 30 marks.

c. Figure on the right of each question indicates marks for the respective question.

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1. a) Define Economics and explain. [5]
   b) Differentiate microeconomics from macroeconomics. [10]
   c) Show the justification for the classification of engineering economics as a separate sub-discipline within the field of economics. [15]

2. a) Explain demand and law of demand with a hypothetical example. [15]
   b) Suppose we are given the following linear demand equation ($Q_d = 100 - 2p$) and supply equation ($Q_s = 20 + 3p$). Find out the value of equilibrium price and equilibrium quantity. [15]

3. a) Elucidate the concept of price elasticity of demand with its different types. [15]
   b) The price of a product increases from Tk 10 to Tk 12. As a result, the quantity demanded decreases from 100 units to 80 units. Calculate the price elasticity of demand (PED) and interpret whether the demand for the product is elastic, inelastic, or unitary elastic. [15]

4. a) Illustrate the concept of opportunity cost using a Production Possibility Frontier (PPF). [15]
   b) What are the three fundamental economic problems that arise due to scarcity in a society? [15]

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Department of Computer Science & Engineering
Level- II Term-II

Midterm Examination, SPRING 2025
Course Code: CSE 223/CSE 201
Course Title: Digital Logic Design
Marks: 90, Time: 1 hour 30 minutes

Notes:

a. Each question carries 30 marks.
b. Figure on the right of each question indicate marks for respective question.
c. Required Postulates and Theorems of Boolean algebra are given in Table 1.

Course Outcomes:

[CO-1]: Remember and understand the number system and Boolean algebra and basic properties of Boolean algebra to simplify simple Boolean functions.
[CO-2]:Understanding and applying the tabulation and Karnaugh map methods for simplifying combinational circuits.

1. a. Define 1's and 2's complements. How are they used in subtraction? Illustrate with binary examples. [CO-1] 18

   b. Subtract the following using 2's complement method: [CO-1] 2x6=12

   - $(10011)_2 - (1101)_2$
   - $(11000)_2 - (10010)_2$

   OR

   b. Prove the following identities using Boolean algebra laws: [CO-1] 2x6=12

   i. $A + A \cdot B = A$
   ii. $A + A' \cdot B = A + B$

2. a. Explain the differences between binary, octal, decimal, and hexadecimal number systems with suitable examples. [CO-1] 06

   b. Convert the following numbers as directed: [CO-1] 4x6=24

   - (Last 4 digit of your ID)$_{10}$ to binary
   - $(1011101)_2$ to decimal
   - $(3F9)_{16}$ to binary
   - $(11011001)_2$ to hexadecimal

OR

b. Simplify the following Boolean functions using Boolean algebra laws: [CO-1] 2x12=24

i. $F(A, B, C) = A \cdot B + A.B' + A'.B$

ii. $F(X, Y, Z) = X \cdot Y + X \cdot Y' \cdot Z + X \cdot Y \cdot Z$

3. a. Use a 4-variable K-map to simplify the function: [CO-2] 15

$$F(w, x, y, z) = \sum_m (0,1,2,5,7,8,9,10)$$

don't-care terms: $d(3,6,11,15)$

b. Simplify the following function using K-Map. [CO-2] 15

$$F(A, B, C, D) = \sum_m (0,1,2,5,6,8,9,10,13,14)$$

Table 1:

1	Postulate 2	(a) X+0 = X	(b) X.1 = X
2	Postulate 5	(a) X+X′ = 1	(b) X. X′ = 0
3	Theorem 1	(a) X+X = X	(b) X.X = X
4	Theorem 2	(a) X+1 = 1	(b) X.0 = 0
5	Postulate 3, Commutative	(a) X+Y = Y+X	(b) XY = YX
6	Theorem 4, Associative	(a) X+(Y+Z) = (X+Y)+Z	(b) X(YZ) = (XY)Z
7	Postulate 4, Distributive	(a) X(Y+Z) = XY+XZ	(b) X+YZ = (X+Y) (X+Z)
8	Theorem 5, De-Morgan	(a) (X + Y)′ = X′ Y′	(b) (XY)′ = X′+Y′
9	Theorem 6, Absorption	(a) X+XY = X	(b) X(X+Y) = X